

Characterizing Gingival Electrical Stimulation Toward Intraoral Texture Display

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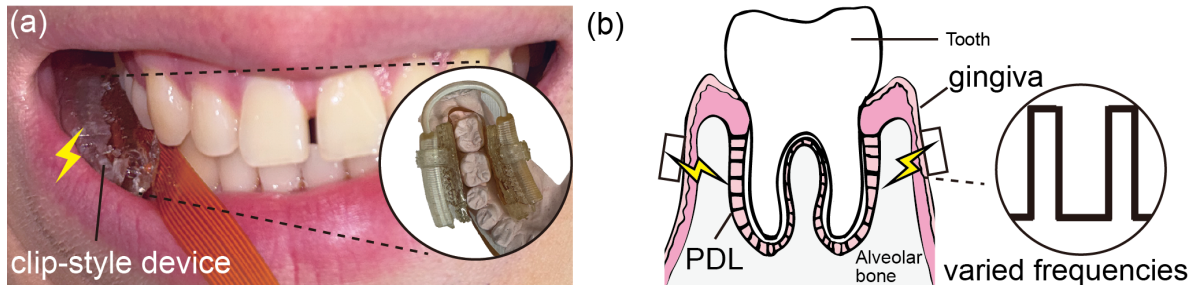


Figure 1: Overview of the gingival electrotactile stimulation approach. (a) A clip-style device delivers electrical stimulation to the buccal and lingual gingiva near the posterior molars without obstructing the occlusal surface. (b) Electrical stimulation is applied across the gingival tissue adjacent to the periodontal ligament (PDL) using varied frequencies (2–500 Hz).

ABSTRACT

Food texture perception during mastication involves kinesthetic and somatosensory feedback. However, the somatosensory contributions of intraoral tissues remain poorly characterized as haptic channels. As a preliminary study toward dynamic texture rendering, we conducted psychophysical profiling of electrotactile stimuli at six frequencies (2–500 Hz) and three electrode locations. Results suggest that the spatial distributions largely overlapped across electrode locations, whereas perceptual qualities were modulated by stimulation frequency. Stimuli between 10–500 Hz tended to be perceived as vibrotactile sensations, while carbonation-like sensations showed a marginal tendency toward higher ratings around 80 Hz.

Index Terms: Oral sensation, electrical stimulation, texture.

1 INTRODUCTION

Haptic perception of food texture during mastication is shaped by both kinesthetic feedback from the masticatory muscles and somatosensory feedback from intraoral tissues such as the periodontal ligament (PDL) and gingiva. The PDL, a thin connective tissue layer between the tooth and alveolar bone, contains dense Ruffini-like mechanoreceptors that respond to tooth loading and convey information related to hardness, and the gingiva contributes tactile cues related to surface properties of food [5].

While muscle kinesthetic feedback and its relation to food texture have been extensively studied, the sensory contribution of periodontal and gingival tissues has rarely been treated as an independent, controllable haptic channel [6, 3]. For intuition, this can be understood as a situation where kinesthetic aspects stand out, while accompanying cutaneous sensations fade into the background.

Previous investigations of PDL and gingival sensation have focused on clinical assessment, such as anesthesia or sensory threshold measurements, rather than on characterizing these sensations from a haptic interaction perspective [7]. Consequently, in interactive system research, oral somatosensory input is often considered together with masticatory motion, making it difficult to disentangle somatosensory effects from kinesthetic feedback.

To address this, we characterize gingival electrical stimulation as an oral haptic channel through a psychophysical study under static jaw conditions, focusing on frequency-dependent perceptual effects. To enable controlled stimulation without interfering with natural tooth contact, we employ a clip-style interface targeting the posterior molars. This work provides a foundation for future interactive systems, including VR, that seek to augment intraoral texture perception during chewing without disrupting natural tooth contact.

2 RELATED WORK

Prior studies have explored food texture presentation through various haptic approaches. Iwata et al. reproduced aspects of food texture using mechanical stimulation and force-based interfaces [3]. Nijima et al. investigated food texture perception by applying electrical stimulation to the masticatory muscles [6]. These approaches primarily focused on kinesthetic feedback during mastication.

In contrast, fewer studies have explicitly targeted intraoral somatosensation from PDL and gingival tissues for sensory modulation. Existing research on oral sensation has mainly examined sensory thresholds or clinical responses, or evaluated the effects of stimulation on mastication, rather than characterizing the qualitative aspects of perceived texture [7, 1]. Although electrical stimulation has been applied to oral tissues, many prior systems relied on a limited set of stimulation parameters, often using a single or narrow frequency range [8, 2]. Since tactile perception varies substantially with parameters such as stimulation frequency, these approaches provide limited insight into generating diverse oral sensations.

3 SYSTEM DESIGN

Clip-Style Electrode Fixture. We developed a compact, clip-style electrode fixture that attaches to the mandibular molars without requiring per-subject customization. The design preserves an unobstructed occlusal plane, allowing direct contact between the upper

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and lower teeth during jaw closure (see Fig. 1a).

Trans-Gingival Stimulation. Flexible flat cable (FFC) electrodes were positioned along the buccal and lingual gingiva of the right posterior molars in three electrode pairs, forming a current path traversing the gingival tissue between each electrode pair. A compliant soft component ensured comfortable contact with the gingiva, while a rigid component provided sufficient retention to maintain stable electrode positioning.

4 USER STUDY

We conducted a psychophysical profiling study ($N = 10$; ages 21–30; 1 female, 9 male) to examine how stimulation frequency and electrode location influence perceived oral sensations. The number of teeth present in the right posterior mandible varied across participants (6 teeth: $n = 1$; 7 teeth: $n = 4$; 8 teeth: $n = 5$).

Participants received stimulation via an electrotactile display at three buccal–lingual gingival electrode pairs across six frequencies (2, 10, 30, 80, 200, and 500 Hz) on the right posterior mandible [4]. Each stimulus was delivered for 5 s. Stimulation amplitude was individually calibrated to a clearly perceivable but non-painful level (1.2–6.0 mA; $M = 3.0$ mA, $SD = 0.8$). Pulse width was fixed at 60 μ s. Polarity was held constant (buccal: cathode; lingual: anode). After each stimulus, participants reported perceived sensation location and rated sensory qualities (e.g., intensity, vibrotactile sensation, pulsation) on a 1–7 Likert scale under a static jaw condition. Because the spatial questionnaire used a seven-tooth layout, analyses of perceived stimulation location were restricted to participants with seven teeth ($n = 4$), whereas sensory-quality ratings were analyzed across all participants ($N = 10$). All procedures were approved by the Institutional Review Board (No. H25018).

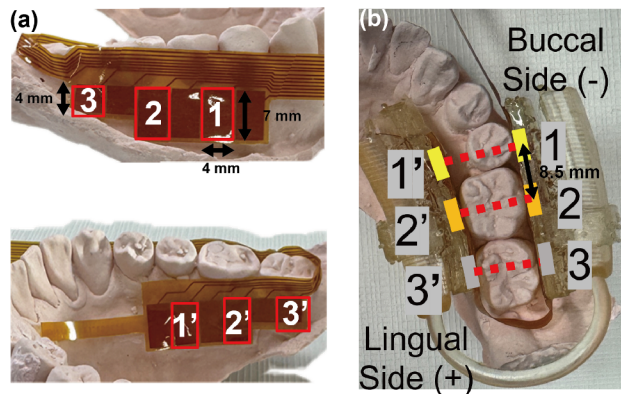


Figure 2: Electrode placement. (a) Side view of the electrodes (b) Positions of the three buccal–lingual electrode pairs on the gingiva.

5 RESULTS AND DISCUSSION

Perceived stimulation locations largely overlapped across the three electrode pairs (spatial reports: $n=4$). With limited individual variability, the sensation was primarily perceived in the gingival surface and interior. Sensations were occasionally felt in unexpected locations, such as the tongue, concurrently. In contrast, sensory-quality ratings ($N = 10$) showed clear frequency dependence. A Friedman test revealed significant effects of frequency on overall intensity at locations 1–1' ($\chi^2(5) = 14.95, p = .011$) and 2–2' ($\chi^2(5) = 15.44, p = .009$), and on vibrotactile sensation at location 2–2' ($\chi^2(5) = 13.97, p = .016$). Consistent with these results, vibrotactile ratings were stronger at 10–500 Hz than at 2 Hz, whereas 2 Hz was predominantly described as pulsation and required higher current to reach a clear percept (Fig. 3). Carbonation-like sensation peaked around 80 Hz, showing a marginal trend at location 3–3'

($p < .10$). Perceived pressure was slightly higher at 200–500 Hz than at lower frequencies, but the main effect of frequency was not significant ($p > .10$).

Sensory Attributes	Overall Intensity	3.5	4.2	4.1	4.4	4.4	4.3
	Vibrotactile	2.8	4.2	4.1	3.8	3.8	3.7
	Pressure	2.1	2.2	2.1	2.1	2.7	2.7
	Pulsation	3.8	2.7	2.3	2.2	2.0	1.8
	Carbonation	1.6	1.9	2.2	2.5	2.0	2.1
		2	10	30	80	200	500
		Frequency (Hz)					

Figure 3: Mean sensory ratings by stimulation frequency. Each cell shows the participants' mean on a 1–7 Likert scale (1 = no sensation, 7 = very strong). Darker colors indicate a stronger sensation.

The dominance of stimulation frequency over electrode location suggests that gingival electrotactile perception is primarily governed by temporal properties rather than spatial properties. The carbonation-like sensation around 80 Hz indicates capabilities beyond simple vibrotactile sensation, highlighting gingival stimulation as a controllable channel for texture-related sensations.

6 CONCLUSION AND FUTURE WORK

This work presents a psychophysical characterization of gingival electrotactile stimulation as an oral haptic channel, focusing on frequency-dependent perceptual effects. The results provide a baseline map of sensations that can be elicited through gingival stimulation. Future work will examine how these frequency-dependent effects generalize to dynamic conditions during chewing and explore their use in interactive scenarios, including VR eating experiences.

7 ACKNOWLEDGEMENT

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